



Technical Reproducibility Validation Report

STRait Razor

Version b618e93

STR / SNP panel

Verification date	2026-08-09
Repository commit	b618e9345ab40f348b504083ae8de2b39abb60fa
Attestation level	Installs
Permalink	https://strhub.app/verified/strait-razor-b618e93
STRhub	https://strhub.app

SCOPE OF THIS REPORT

Independent automated verification of reproducible execution. It confirms the tool installs, runs end-to-end, and produces structurally valid output in a standardized environment.

This is **not** an analytical validation. Genotype accuracy, concordance, forensic suitability, and regulatory compliance are out of scope.

WHAT WAS VERIFIED

- Source available
- Installation successful
- End-to-end execution
- Output generated

NOT VERIFIED

- Genotype accuracy
- Concordance
- Forensic validity
- Regulatory compliance



1. Executive Summary

PURPOSE	Verify that the tool installs, runs end-to-end, and produces structurally valid output.
RESULT	2/5 gates passed. Installs
ATTESTATION LEVEL	Installs
SCOPE	Installation, execution, and output structure verification.
NOT EVALUATED	Genotype accuracy · Concordance · Forensic validity · Regulatory compliance

2. Reproducibility Metadata

Tool	STRait Razor
Version	b618e93
Repository	https://github.com/Ahhgust/STRaitRazor
Commit (pinned)	b618e9345ab40f348b504083ae8de2b39abb60fa
Container OS	ubuntu-22.04
Marker panel	STR / SNP panel
Verified on	2026-08-09 22:41:38 UTC
CI run	https://github.com/Tfronta/strhub-verified/actions/runs/31340014066

3. Exact Run Command

The following command was executed verbatim in the CI environment. Replacing the input paths under /data/in with your own inputs reproduces the verified run.

```
# Environment: ubuntu-22.04 · STRait Razor b618e93 · commit b618e93
$ ./str8rzt \
-c \
https://github.com/Ahhgust/STRaitRazor/blob/master/ForenSeqv1.27.config \
/data/in/sample.fastq \
> \
/data/out/allsequences.txt
```

4. Verification Gates

Available	The pinned public source exists	PASS
Installs	The environment builds from source	PASS
Runs	Executes end-to-end without crashing	—
Expected IO	Produces a non-empty file in the declared format	—

Output Structure Validation	Output structure matches expected genotype-bearing format	—
Execution log	strait-razor-b618e93.log-external.txt	

5. Out of Scope

This report does not evaluate any of the following:

- Genotype correctness or accuracy
- Concordance against known truth sets
- Sensitivity, specificity, or stutter performance
- Allele calling accuracy or forensic casework suitability
- Regulatory compliance or ISO accreditation
- Multi-laboratory or multi-dataset reproducibility

6. Output Content Evidence

Output file	allsequences.txt
Format	TSV
Sequence records	0
Distinct STR loci	0
Total reads	0
Max depth (single locus)	0

7. Verification Data

This tool does not include its own demo or test data. STRhub ran the verification using the public reference dataset listed below.

Dataset	NIST mds2-2157, Illumina STR (ForenSeq slice, donor NTD01)
Source	https://data.nist.gov/od/id/mds2-2157
License	research / training / education only (per NIST); not for donor identification or database searching
Ref. genome	—
Loci tested	—

8. Verification Matrix

PASS	Reference dataset	NIST mds2-2157, Illumina STR (ForenSeq slice, donor NTD01)
	README check (5/5)	Advisory: all items present
	Install / environment setup	Present
	Run command	Present
	Expected input format	Present
	Produced output format	Present
	Dependencies / versions	Present

9. Limitations

- Single reference dataset per input type
- Single containerized environment (Docker / ubuntu-22.04)
- No truth-set comparison or ground-truth genotypes
- No accuracy or concordance assessment
- No forensic validation of results
- Short-read limitations apply (very long STR alleles may not span reads)

10. Scope and Disclaimers

Executed end-to-end in the stated environment with output in the expected format. Concerns reproducible execution only; no claim of accuracy, casework fitness, or regulatory validation.

Each result is a dated snapshot, verified on the tool's public repository at a pinned commit. STRhub does not store tool source code.

11. Conclusion

◆ Runs end-to-end

STRait Razor b618e93 installs and executes without error in a clean ubuntu-22.04 environment at the pinned commit. All 2 verification gates were passed.

◆ Produces valid output

The tool generated a structurally valid TSV output file with genotype calls across 0 target forensic STR loci, with a maximum read depth of 0 at —.

◆ No bundled demo data

STRait Razor does not include its own test or demo dataset. STRhub Verified supplied NIST mds2-2157, Illumina STR (ForenSeq slice, donor NTD01) as the reference input for this verification run. Users replicating this result should use the same or equivalent publicly available reference material.

◆ Reproducibility statement

The exact command reported in Section 3, executed at the pinned commit in the described environment, is sufficient to reproduce this verification result independently.